



Department of Computer Science

AI5004 – Advanced Topics in Generative AI

Fall 2026

Instructor Name: Maryam Bashir

Email address: maryam.bashir@nu.edu.pk

Office Location/Number: New Building, 11

Office Hours: Monday, Wednesday 11:30 AM to 1:00 PM

Course Information

Program: MSDS

Credit Hours: 3

Type: Elective

Pre-requisites: AI

Class Meeting Time: MDS-3A: Thursday 8:30 AM to 11:30 AM

Class Venue: F-204

Course Description:

This course provides an in-depth understanding of Generative AI, exploring its foundational models, advanced architectures, and practical applications. Students will begin with generative models like GANs and diffusion models, followed by transformers and large language models (LLMs), including their architectures, training techniques, and industry applications. The course emphasizes hands-on learning with fine-tuning techniques (LoRA, QLoRA, RLHF), optimization methods, and evaluation metrics for generative models. Additionally, students will explore prompt engineering strategies, retrieval-augmented generation (RAG), and AI agents such as LangChain. Through this course, students will gain the knowledge and skills required to design, evaluate, and apply generative AI systems effectively across various domains.

Course Learning Outcomes (CLOs):

At the end of the course students will be able to:

- **Explain foundational principles of Generative AI models**, including GANs, diffusion models, and transformers, and their roles in generating diverse data types.
- **Analyze the architectures and pre-training methodologies of large language models (LLMs)** and their evolution from traditional NLP techniques.
- **Evaluate the optimization methods and evaluation metrics** used for assessing the quality and performance of generative models.
- **Discuss advanced topics such as prompt engineering, retrieval-augmented generation (RAG), and fine-tuning techniques** in the context of their theoretical foundations and applications.

Course Textbook

Generative Deep Learning, second edition, Teaching Machines to Paint, Write, Compose, and Play-
David Foster Foreword by Karl Friston

Reference Books

1. Generative AI in Practice by Bernard Marr
2. RAG-Driven Generative AI: Build Custom Retrieval Augmented Generation Pipelines with LlamaIndex, Deep Lake, and Pinecone Book by Denis Rothman

Additional references and research papers related to the course:

Weekly Schedule

Week	Topics Covered	Text Book Chapter
1	Overview of Generative AI: Definition, Key Concepts, and Historical Development, Applications of Generative AI in Text, Images, and Beyond, Comparison of Generative vs. Discriminative Models Generative Modeling Framework, Autoencoders, VAE	1, 3
2	Introduction to GANs: Architecture (Generator and Discriminator) Minimax Game and Loss Functions	3
3	Diffusion Models: Introduction and Theory Noise Scheduling and Denoising Diffusion Probabilistic Models (DDPMs) Applications in Text-to-Image and Video Generation	4, 8
4	Multimodality: CLIP Transformer Architecture: Attention Mechanism, Encoder-Decoder Framework Applications in Sequence-to-Sequence Tasks (e.g., Translation, Summarization), Variants of Transformers: Vision Transformers (ViT), Efficient Transformers, Improvements over Vanilla Transformers	9
5	Stable Diffusion, Architecture, Training, Sampling & Inference Prompting and Text-to-Image Generation	
Midterm 1		
6	LLM Architecture: Mistral	
7	Mixture of Experts (MoE) Architecture and Application	
8	Fine Tuning Techniques: Task-Specific Fine-Tuning: Adapting Pre-Trained Models for New Tasks Parameter-Efficient Fine-Tuning (PEFT) Diverse Optimization Techniques including Prefix Tuning or Adapter Layers, Prompt Tuning, Quantization: Reducing Model Size Without Performance Loss (LoRA, QLoRA) Introduction to LoRA and QLoRA Implementation of LoRA and QLoRA in Practice	
9	Introduction to Retrieval-Augmented Generation (RAG) RAG Pipelines	
10	GraphRAG with Research Paper, Corrective RAG with Research Paper, Handling Hallucinations	
Midterm 2		
11	Introduction to AI Agents: LangChain and LangGraph Building Dynamic Workflows with LangChain Hands-On: Designing a Simple AI Agent Workflow	

12	Presentations	
13	Presentations	
14	Presentations	

Grading Scheme

The grading scheme will be **absolute grading** scheme

Tentative Grading Criteria

1. 4 Assignments (20%)
2. 3 Quizzes (9%)
3. Midterm Exams (20%)
4. Final Exam (46%)
5. Presentation (5%)

Course Policies

1. Quizzes will be announced.
2. No makeup for missed quiz or assignment.
3. 80% attendance

Plagiarism in Assignments

In writing up your assignments and in answering questions in exams, be as clear, precise, and concise as you can. **Understandability will be an important factor in grading.**

Academic Integrity: All work MUST be done individually. Any copying of work from other person(s) or source(s) (e.g. the Internet) will automatically result in at least an F grade in the course. It does not matter whether the copying is done in an assignment, quiz, midterm exam, or final exam, it will be considered equally significant.